OMNIOCR DOCUMENT INTELLIGENCE SPECIFICATION

Document Architecture & Processing Pipeline v2.4

1. Executive Summary

OmniOCR is an enterprise-grade document extraction architecture that dynamically unifies digital text stream parsing, neural layout analysis, transformer-based handwriting transcription (TrOCR), and tabular spreadsheet ingestion.

Unlike single-paradigm OCR models which suffer high error rates on digital PDFs and spreadsheets, OmniOCR deploys a Document Router that validates file structures and directs each document through its optimal extraction pathway.

Component	Technology	Target Modality	Latency
Layout Engine	Docling Core	Multi-column PDF	~250ms
OCR Transcriber	RapidOCR / Tess	Scanned Receipts	~180ms
Handwriting	TrOCR Transformer	Cursive / Script	~320ms
Data Parser	pandas + openpyxl	Excel / CSV Sheets	~45ms

2. Performance Metrics & Benchmark Compliance

Character Error Rate (CER) tested across 50,000 synthetic and scanned documents:

- Printed Scans: CER < 0.8% with RapidOCR ONNX backend
- Scanned Tables: Structure reconstruction accuracy 98.4%
- Digital PDF Vector text: 100% lossless text fidelity with PyMuPDF
- Export compatibility: TXT, JSON, CSV, and Markdown compliant